```
!pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

```
import numpy as np
print(np.__version__)
```

```
2.0.2
```

```
arr=np.array([1,2,3,4,5,6,7])
print(arr)
```

```
[1 2 3 4 5 6 7]
```

```
a=np.array([[1,2,3],[4,5,6]])
print(a)
```

```
[[1 2 3]
 [4 5 6]]
```

```
b=np.array([[[1,2,3],[4,5,6]],[[7,8,9],[1,2,3]]])
print(b)
```

```
[[[1 2 3]
  [4 5 6]]

 [[7 8 9]
  [1 2 3]]]
```

```
d=np.zeros([3,3])
print(d)
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
d=np.ones([3,3])
print(d)
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
d=np.full([3,3],8)
print(d)
```

```
[[8 8 8]
 [8 8 8]
 [8 8 8]]
```

```
def greet():
  """
  this is for testing doc string
  """
  print("testing function")
print(greet())
```

```
testing function
None
```

```
f=np.arange(2,10,3)
print(f)
```

```
[2 5 8]
```

```
np.linspace(0,1,5)
```

```
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
```

```
#uniform random
np.random.rand(3,3)
```

```
array([[0.34456506, 0.29706676, 0.51534777],
       [0.24610631, 0.38781733, 0.91644552],
       [0.2791709 , 0.95203621, 0.82129525]])
```

```
#random integers
np.random.randint(0,10,size=(3,3))
```

```
array([[9, 2, 1],
       [2, 2, 0],
       [3, 1, 2]])
```

```
#intialize grabage value
np.empty((2,4))
```

```
array([[1.73136205e-315, 0.00000000e+000, 0.00000000e+000,
        0.00000000e+000],
       [0.00000000e+000, 0.00000000e+000, 0.00000000e+000,
        0.00000000e+000]])
```

```
arr=np.array([["a","b","c","d","d","e"]])
print(arr.shape)
print(arr.ndim)
print(arr.size)
print(arr.dtype)
print(arr.itemsize)
print(arr.nbytes)
```

```
(1, 6)
2
6
<U1
4
24
```

```
a=np.array([10,20,30,40,50])
print(a[0])
print(a[2])
print(a[-1])
print(a[1:4:2])
print(a[::2])
```

```
10
30
50
[20 40]
[10 30 50]
```

```
#slicing
print(arr[1:4])
print(arr[1:3])
print(arr[::2])
print(arr[::-1])
```

```
[2 3 4]
[2 3]
[1 3 5 7]
[7 6 5 4 3 2 1]
```

```
#2d indexing
m=np.array([[1,2,3],
          [2,3,4],
          [5,6,7]])
print(m[0,0])
print(m[1,2])
#row aslicing
print(m[0,:])
#column slicing
print(m[:,1])
#sub-matrix
print(m[0:2,1:3])
```

```
1
4
[1 2 3]
[2 3 6]
[[2 3]
 [3 4]]
```

```
a=np.array([10,25,8,42,17])
#boolen mask
mask=a>15
print(a[mask])
#short hand
print(a[a>15])
print(a[[0,3]])
```

```
[42 17]
[42 17]
[10 42]
```

```
import numpy as np
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
print(a-b)
print(a*b)
print(a/b)
print(a**2)
print(a+10)
```

```
[5 7 9]
[-3 -3 -3]
[ 4 10 18]
[0.25 0.4  0.5 ]
[1 4 9]
[11 12 13]
```

```
a=np.array([1.0,4.0,9.0,12.0])
np.ceil(a)
np.floor(a)
print(a)
```

```
[ 1.  4.  9. 12.]
```

```
import numpy as np
a=np.arange([2,20]])
```

```
[ 2 20]
```